Machine Learning

Problem Set 02

Instructions. If you think any question is too difficult, you can give it up strategically. Please arrange your time according to your own schedule.

Question 1.

Compare the three variance reduction algorithms - SAG, SVRG and SAGA from different aspects (e.g.gradient estimation formula, convergence speed, stochastic gradient biased or not, and other advantages and disadvantages).

Question 2.

In our classroom discussions regarding Mirror Descent (MD), we have become well-acquainted with the update procedure, where the position vector $\mathbf{x}_{t+1}$ is updated as follows:

$$\mathbf{x}_{t+1} := \operatorname*{argmin}_{\mathbf{y}} \left(V_{\mathbf{x}_t}(\mathbf{y}) + \langle \alpha \cdot \nabla f(\mathbf{x}_t), \mathbf{y} - \mathbf{x}_t \rangle \right). \tag{1}$$

$V_{\mathbf{x}}(\mathbf{y}) = R(\mathbf{y}) - \langle \nabla R(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle - R(\mathbf{x})$ stands for the Bregman Divergence.

Now, we introduce a new algorithm that, at each step, updates $\mathbf{x}_{t+1}$ as follows:

$$\mathbf{x}_{t+1} := \operatorname*{argmin}_{\mathbf{y}} \left(\alpha \sum_{s=1}^{t} \langle \mathbf{y}, \nabla f(\mathbf{x}_s) \rangle + R(\mathbf{y}) \right), \tag{2}$$

where $R(\mathbf{y}) : \mathbb{R}^d \to \bar{\mathbb{R}}$ is a strongly convex function that appears in $V_{\mathbf{x}}(\mathbf{y})$.

Please demonstrate the equivalence of this algorithm to MD. Both algorithm takes $\mathbf{x}_1 = \operatorname{argmin}_{\mathbf{y}} R(\mathbf{y})$ as initial point.

Question 3.

Let $f(x) = \frac{1}{n}\sum_{i=1}^{n} f_i(x)$ represent the finite sum of n L-smooth and convex functions $f_i(x)$.

For each epoch $s \in [S]$ and step $t \in \{0, 1, \ldots, m_s - 1\}$, where m_s doubles between every consecutive 2 epochs, let i_t^s represent the randomly selected index $i \in [n]$, and ξ_t^s be the stochastic gradient, given by $\xi_t^s = \nabla f_{i_t^s}(x_t^s) - \nabla f_{i_t^s}(\tilde{x}^{s-1}) + \nabla f(\tilde{x}^{s-1})$. Here, x_t^s denotes the point at step t in epoch s, and $\tilde{x}^{s-1}$ is the average point from the previous epoch.

We define $F(x) \stackrel{\text{def}}{=} f(x) + \Psi(x) \stackrel{\text{def}}{=} \frac{1}{n}\sum_{i=1}^{n} f_i(x) + \Psi(x)$, where $\Psi(x)$ is a relatively simple, though possibly non-differentiable, convex function known as the *proximal function*. Finally, let x^* be the minimizer of $F(x)$.

Prove:

(a) The expected squared difference between the stochastic gradient ξ_t^s and the true gradient $\nabla f(x_t^s)$ is bounded by:

$$\mathbb{E}_{i_t^s}\Big[\|\xi_t^s - \nabla f(x_t^s)\|^2\Big] \leq 2 \cdot \mathbb{E}_{i_t^s}\Big[\|\nabla f_{i_t^s}(x_t^s) - \nabla f_{i_t^s}(x^*)\|^2 + \|\nabla f_{i_t^s}(\tilde{x}^{s-1}) - \nabla f_{i_t^s}(x^*)\|^2\Big].$$

(b) $\mathbb{E}_{i_t^s}\left[\|\xi_t^s - \nabla f(x_t^s)\|^2\right]$ is further bounded by:

$$\mathbb{E}_{i_t^s}\Big[\|\xi_t^s - \nabla f(x_t^s)\|^2\Big] \leq 4L \cdot \Big(F(x_t^s) - F(x^*) + F(\tilde{x}^{s-1}) - F(x^*)\Big).$$

Hint: For a convex and L-smooth function f, it holds that

$$f(y) \geq f(x) + \langle \nabla f(x), y - x\rangle + \frac{1}{2L}\|\nabla f(x) - \nabla f(y)\|^2.$$

But you **need to prove the inequality** before using it.

Here are some key facts about general convex functions (perhaps non-differentiable) and subgradients, which you can use without proof:

A vector g is defined as a **subgradient** of the function h at a point x_0 if, for any point x, the following inequality holds:

$$h(x) \geq h(x_0) + \langle g, x - x_0\rangle.$$

The set of all subgradients of h at x_0 is denoted by $\partial h(x_0)$. If h is a convex function, then for any point x, $\partial h(x)$ is non-empty. Further, if $x^* \in \operatorname{argmin}_x h(x)$, then $0 \in \partial h(x^*)$.

Question 4.

In this question, we will learn multiplicative weight update (MWU) method, and see why it is a special case of Mirror Descent, as well as why sometimes MD can be helpful.

Given an action space $\Delta_n \triangleq \{x \in [0,1]^n : \|x\|_1 = 1\}$, where each vector in Δ_n is an action (which is also a probability vector). We consider the online setting, where we have to play an action $x_t \in \Delta_n$ in step $t \in [T]$. In the t-th step, **after** we picked our x_t, we will get a loss vector $\ell_t \in \mathbb{R}^n$ that may depend on x_t, which incurs loss $x_t^T \ell_t$. In this question, if $x \in \Delta_n$, we use x^i to denote its i-th coordinate. If $x_t \in \Delta_n$, we use x_t^i to denote its i-th coordinate.

In the MWU setting, we assume $\|\ell_t\|_\infty \le \rho$ for all $t \in [T]$. We are interested in minimizing the average regret, which is defined as:

$$\frac{1}{T}\left(\sum_{t=1}^{T} x_t^T \ell_t - \min_{x_*} \sum_{t=1}^{T} x_*^T \ell_t\right)$$

That is, what is the gap between our strategy and the strategy that picks the optimal fixed x_*. Now, please answer the following questions:

(a) Why can't we directly use gradient descent (GD) here? That is, we maintain our x vector as our weight vector. In every step, we compute the gradient of the loss function at t-th step (in this case, ℓ_t), and apply the gradient step.

(b) Consider the projected gradient descent algorithm (PGD): $x_{t+1} = \Pi_{\Delta_n}(x_t - \eta \ell_t)$, where $\Pi_{\Delta_n}(x) = \text{argmin}_{y \in \Delta_n} \|x - y\|_2$. Please show that this algorithm is equivalent to mirror descent using the standard ℓ_2-norm as $V_x(y)$. Moreover, prove that to get average regret of ϵ, we need $O(\frac{\rho^2 n}{\epsilon^2})$ steps.

(c) Consider the MWU method described in Algorithm 1. Please show that, if $w(x) = \sum_{i \in [n]} x^i \log x^i$, where $x \in \Delta_n$, the corresponding Bregman divergence is defined as $V_x(y) = \sum_{i \in [n]} y^i \log \frac{y^i}{x^i}$. Then, this MWU algorithm is equivalent to mirror descent using $V_x(y)$.

(d) Please prove that to get average regret of ϵ, we need $O(\frac{\rho^2 \log n}{\epsilon^2})$ steps. Therefore, by picking a better $V_x(y)$, we can get better convergence rate. Hint: in this case, in the analysis of mirror descent, we will get the following guarantee because we use a different norm in Bregman divergence:

$$\alpha(f(x_k) - f(u)) \le \frac{\alpha^2}{2}\|\nabla f(x_k)\|_\infty^2 + V_{x_k}(u) - V_{x_{k+1}}(u)$$

Notice that we have ℓ_∞-norm here. This is because the proof for deriving that line in ppt (i.e., $(a-b)^2 \ge 0$) is due to Holder's inequality. Moreover, the Bregman divergence used here is strongly convex with respect to ℓ_1 norm. Please also include these steps in your proof.

Algorithm 1 Multiplicative Weights Update (MWU)

1: Initialize $x_1 = \frac{1}{n}\mathbb{1}$ (the uniform distribution).
2: **for** each iteration $t = 1, 2, \ldots, T$, **do**
3: Obtaining $\ell_t \in \mathbb{R}^n$ from the oracle.
4: Let $w_{t+1} \in \mathbb{R}^n$ and $x_{t+1} \in \Delta_n$ be defined as

$$w_{t+1}^i := x_t^i e^{-\eta \ell_t^i}, \quad x_{t+1}^i := \frac{w_{t+1}^i}{\sum_{j=1}^n w_{t+1}^j}, \quad i \in [n]$$

5: **end for**